

Shape Optimization of Stress-Relieving Auxiliary Holes in a Plate with Central Hole

A Gradient-Based Multi-Start Optimization Approach

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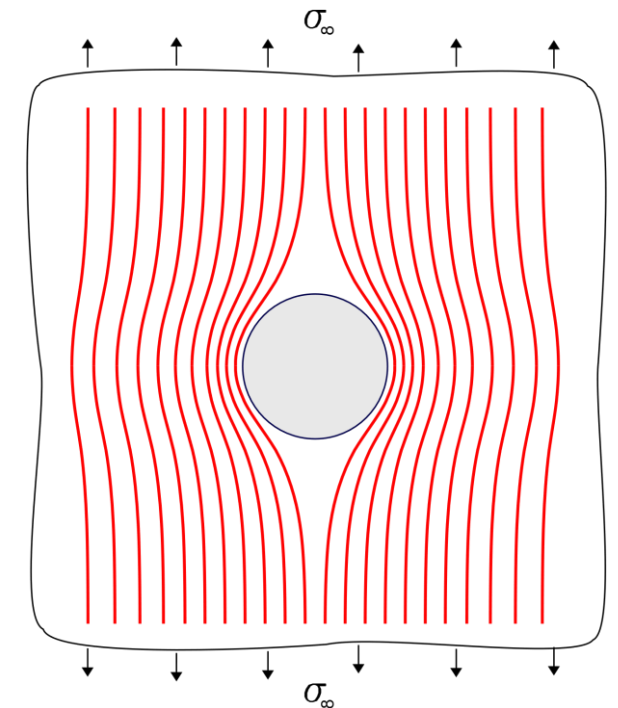
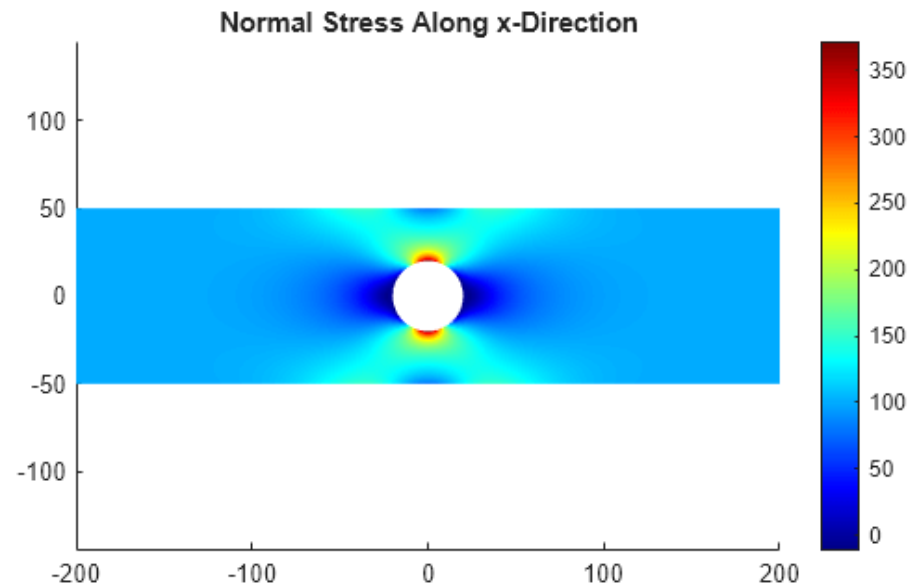
ME7761 Optimum Design of Machines and Structures

Course Instructor: Anthony Luscher

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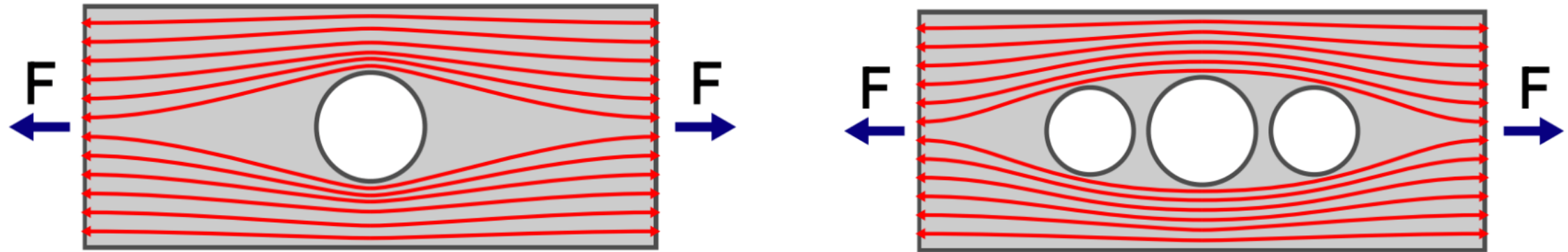
Motivation and Problem

- Holes, notches, and geometric discontinuities causes stress concentration
- Stress concentration factor K_t for a circular hole in infinite plate: $K_t \approx 3.0$ ([Kirsch solution, 1898](#))
- Fatigue failures initiate at stress concentration sites
- Can we reduce stress concentration by adding **MORE** holes?

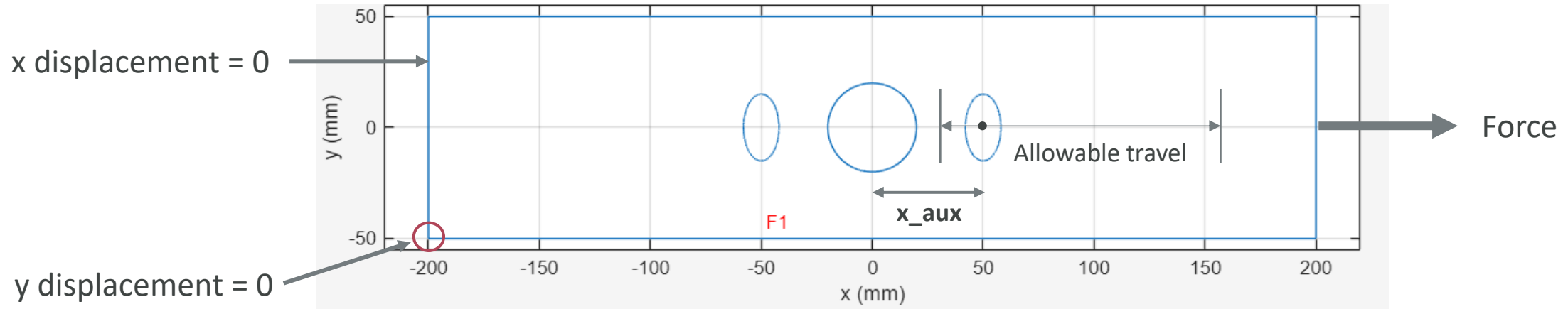


Stress-Relief Through Auxiliary Holes

- Auxiliary holes intercept and redistribute stress flow
- Trade-off: auxiliary holes create their own (smaller) stress concentrations



Problem Geometry



Parameter	Symbol	Value
Plate dimensions	$2L \times 2W$	400×100 mm
Central hole radius	R_c	20 mm
Material	Steel	$E = 200$ GPa, $\nu = 0.25$
Loading	σ_0	100 MPa (uniaxial tension)

Variable	Description	Bounds
x_{aux}	Distance from center	[38, 150] mm
a	Semi-major axis	[6, 15] mm
b	Semi-minor axis	[6, 15] mm
θ	Rotation angle	$[-90^\circ, 90^\circ]$

Note: Ellipse are symmetrical to each other

Optimization Problem Formulation

- **Minimize:** Maximum von Mises stress anywhere in the plate

$$\begin{aligned} \min_{\mathbf{x}} f(\mathbf{x}) &= \max_{(x,y) \in \Omega} \sigma_{vM}(x,y;\mathbf{x}) \\ \text{subject to } g_i(\mathbf{x}) &\leq 0, i = 1, \dots, 4 \\ \mathbf{x}^L &\leq \mathbf{x} \leq \mathbf{x}^U \end{aligned}$$

- Where Ω is the plate domain (excluding holes)
- $\mathbf{x}$ is box bounds
- σ_{vM} is the von Mises equivalent stress

$$\sigma_{vM} = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\tau_{xy}^2}$$

Geometric Constraints (Inequality):

g_1 : Non-overlapping with central hole ≤ 0

g_2 : Right edge clearance (x) ≤ 0

g_3 : Top/bottom edge clearance (y) ≤ 0

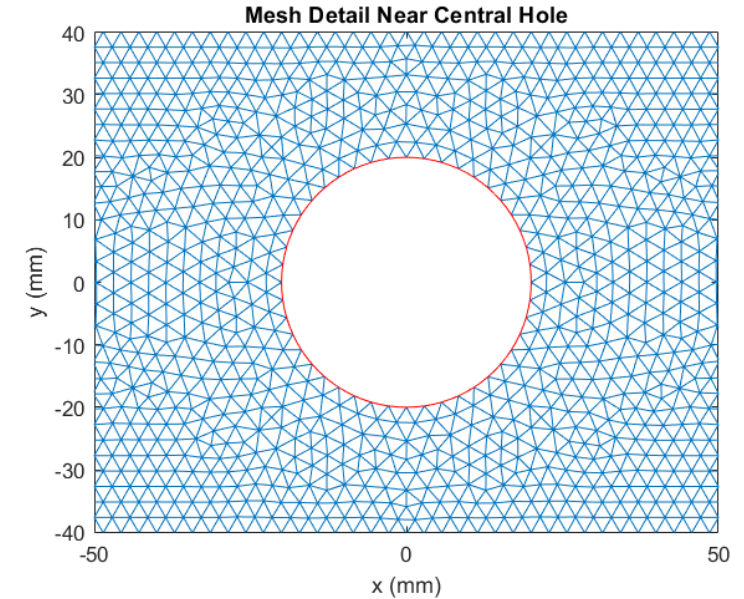
g_4 : Aspect ratio limit ≤ 0

Finite Element Analysis & Baseline Reference

FEA Stats

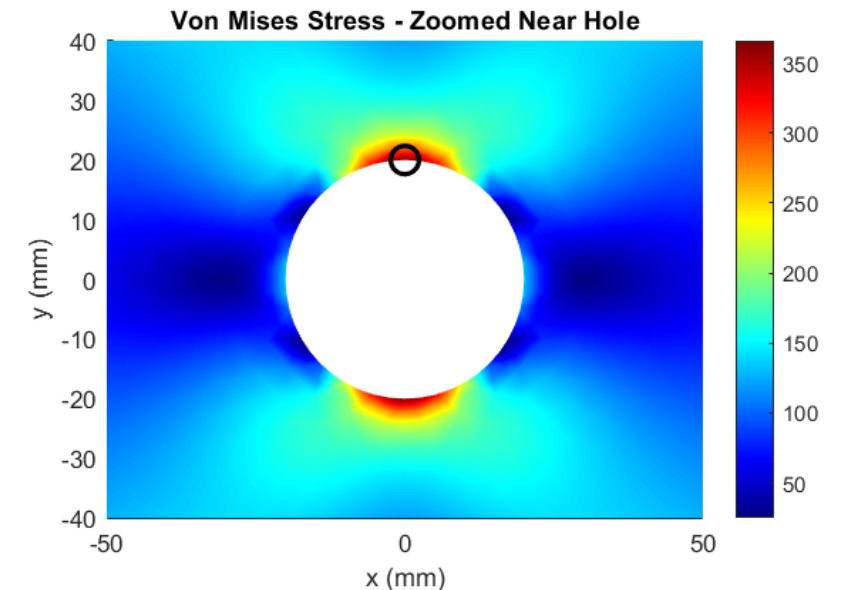
Parameter	Value
Mesh type	Adaptive triangular
Elements	~38,000
Nodes	~77,000
Solve time	~0.3-0.5 s per evaluation

- Performed basic mesh convergence study
- Mesh number looks excessive but had to be this large to simulate far-field stress (very long plate, figure is zoomed in here)



Baseline Analysis

Metric	Value
Max σ_{vM}	360.28 MPa
Stress Concentration Factor	$K_t = 3.60$
Location	Top of central hole (0, 20) mm



Solution Approach

- Analysis: 2D Plane-Stress FEA ([MATLAB PDE Toolbox](#))
- Optimizer: fmincon with SQP algorithm
- Gradients: Finite difference (central)
- Multi-start: Latin Hypercube Sampling (248 points)
- Parallel Processing using 8 cores

```
options = optimoptions('fmincon', ...  
    'Algorithm', 'sqp', ...  
    'Display', 'iter', ...  
    'MaxIterations', 200, ...  
    'MaxFunctionEvaluations', 1200, ...  
    'OptimalityTolerance', 1e-4, ...  
    'StepTolerance', 1e-4, ...  
    'ConstraintTolerance', 1e-4, ...  
    'FiniteDifferenceType', 'central', ...  
    'UseParallel', true);
```

Why Multi-Start?

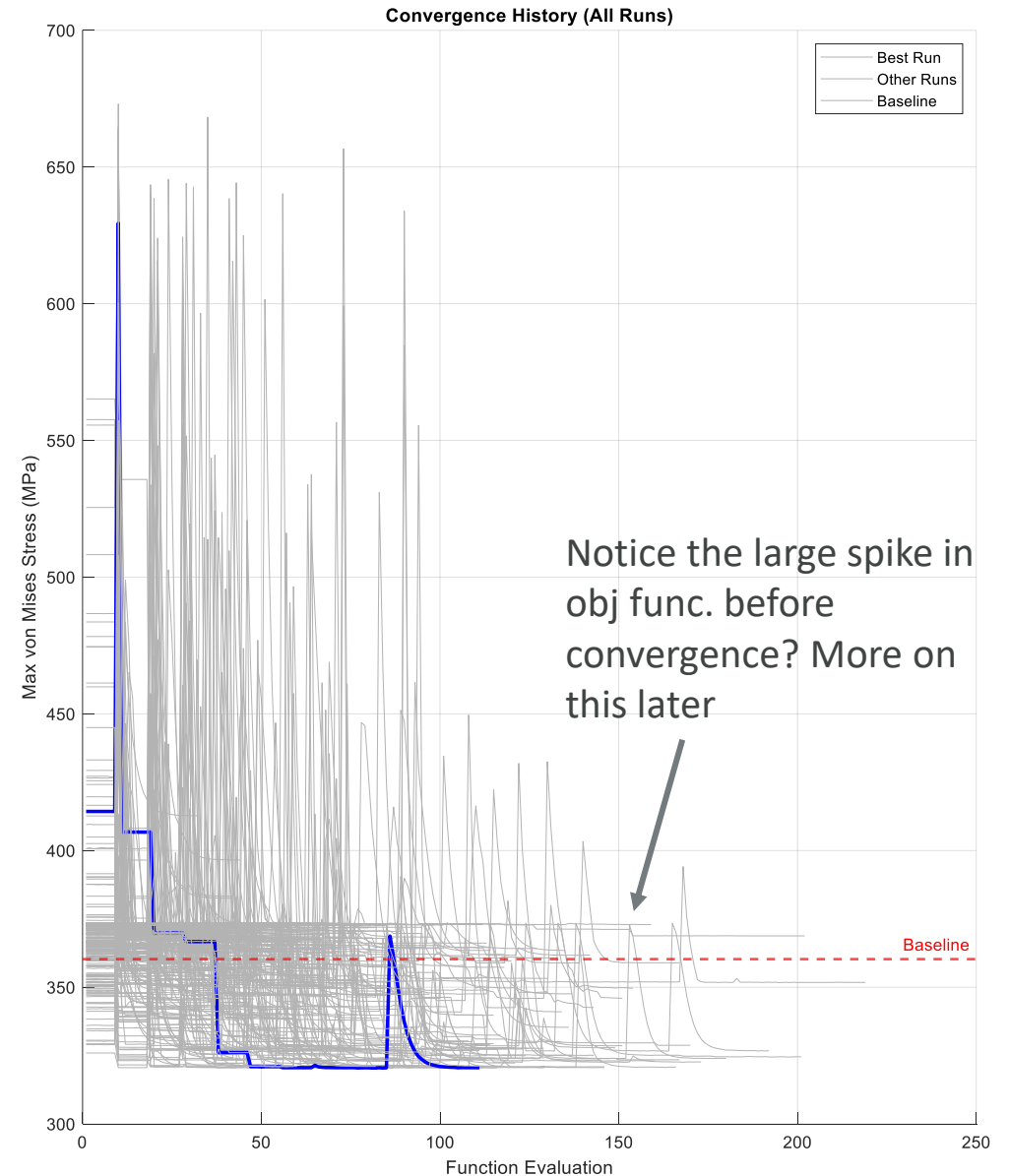
- Stress field is complex
- Multiple local minima expected
- Global optimum not guaranteed, but robust exploration

Multi-Start Optimization Convergence

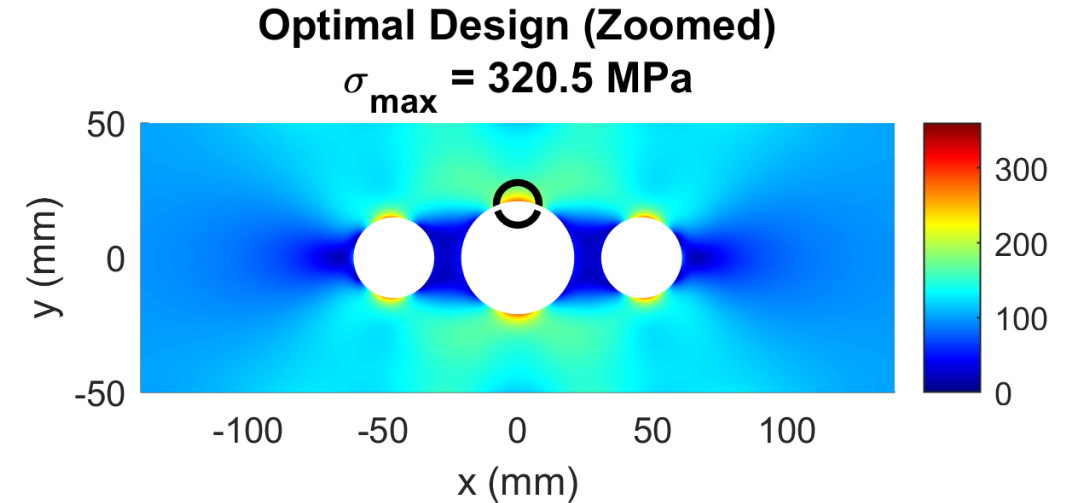
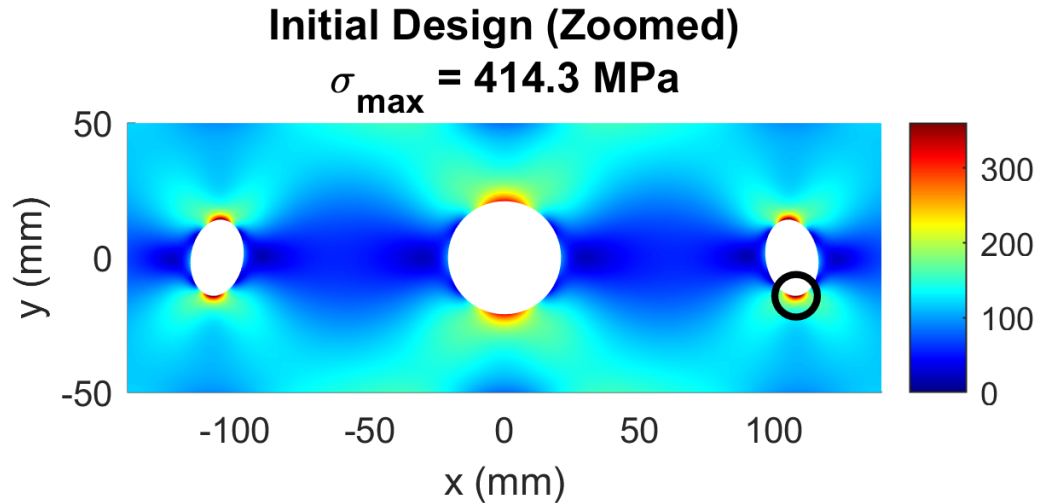
Key observation:

- Wide spread in final objectives (320-413 MPa)
- Many runs converge to local minima
- Best run achieve 11.1% reduction below baseline

Metric	Value
Multi-start runs	248
Total FEA (func) evaluations	19,864
Average per run	~80 evaluations
Best run found	Run #199
Total time	2.6 hours
CPU	Intel i9-12900K 3.2 GHz



Optimization Results



Optimal Parameters:

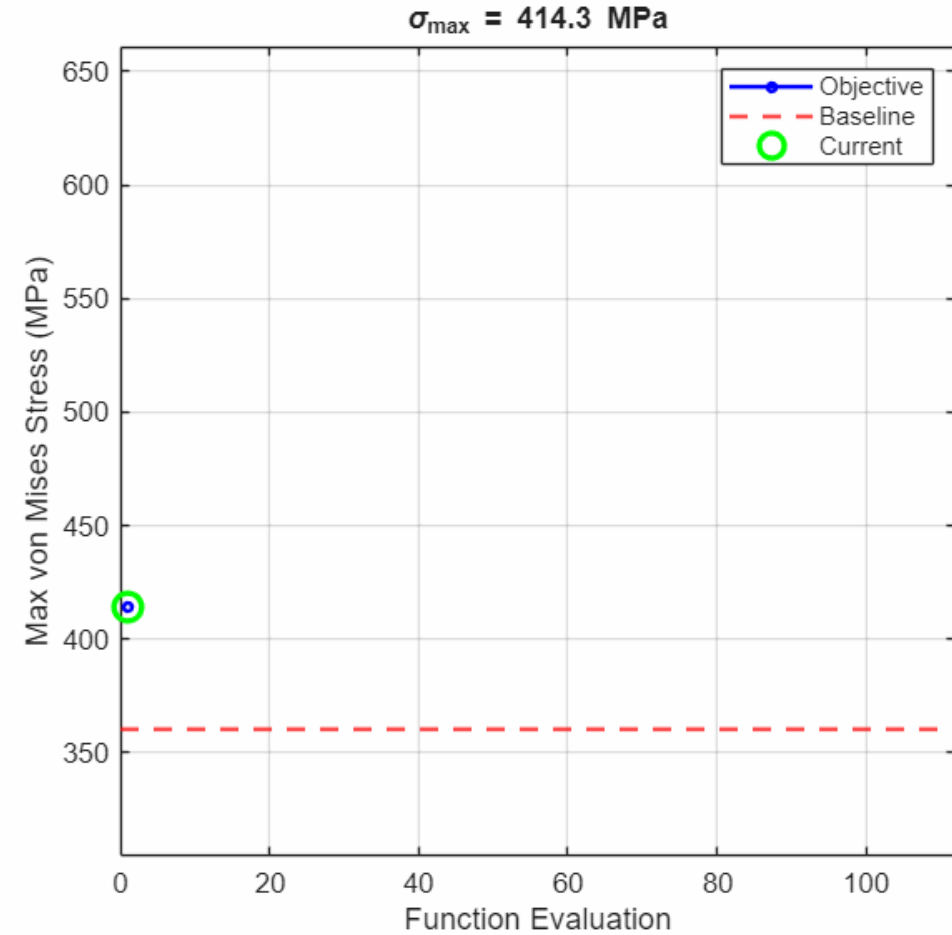
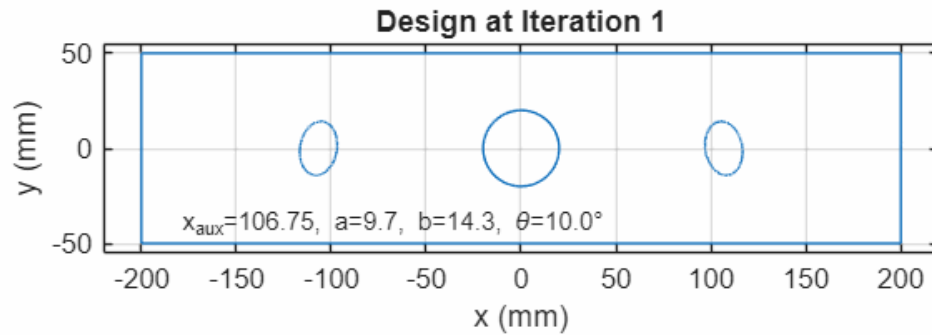
- $x_{\text{aux}} = 46.0 \text{ mm}$ ($2.3 \times$ central hole radius)
- $a = b = 15.0 \text{ mm}$ (**circular**)
- $\theta = -35.8^\circ$ (irrelevant for circle)

Optimizer converged to **circular** auxiliary holes at maximum allowed size

Metric	Baseline	Optimized	Change
Max σ_{vM}	360.3 MPa	320.5 MPa	-11.1%
K_t	3.60	3.20	

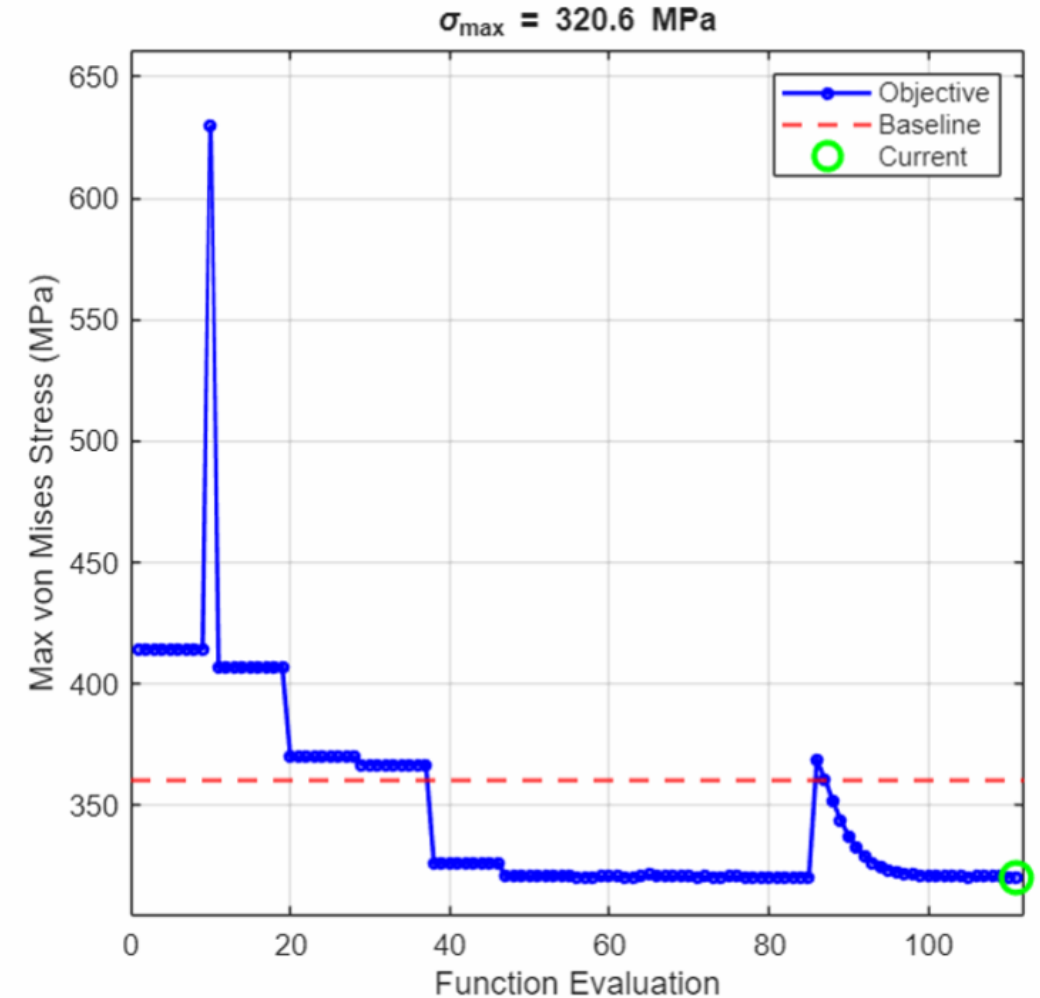
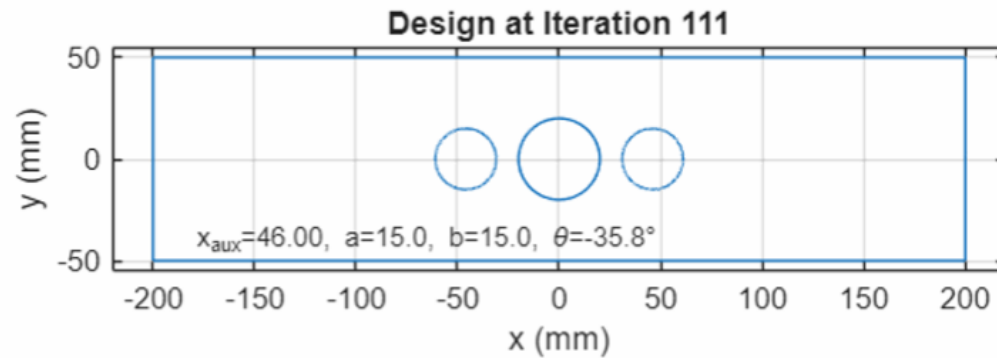
Animation of Rank-1 Solution

Run 199 - Frame 1/40



Rank 1 – Final Solution

Run 199 - Frame 40/40

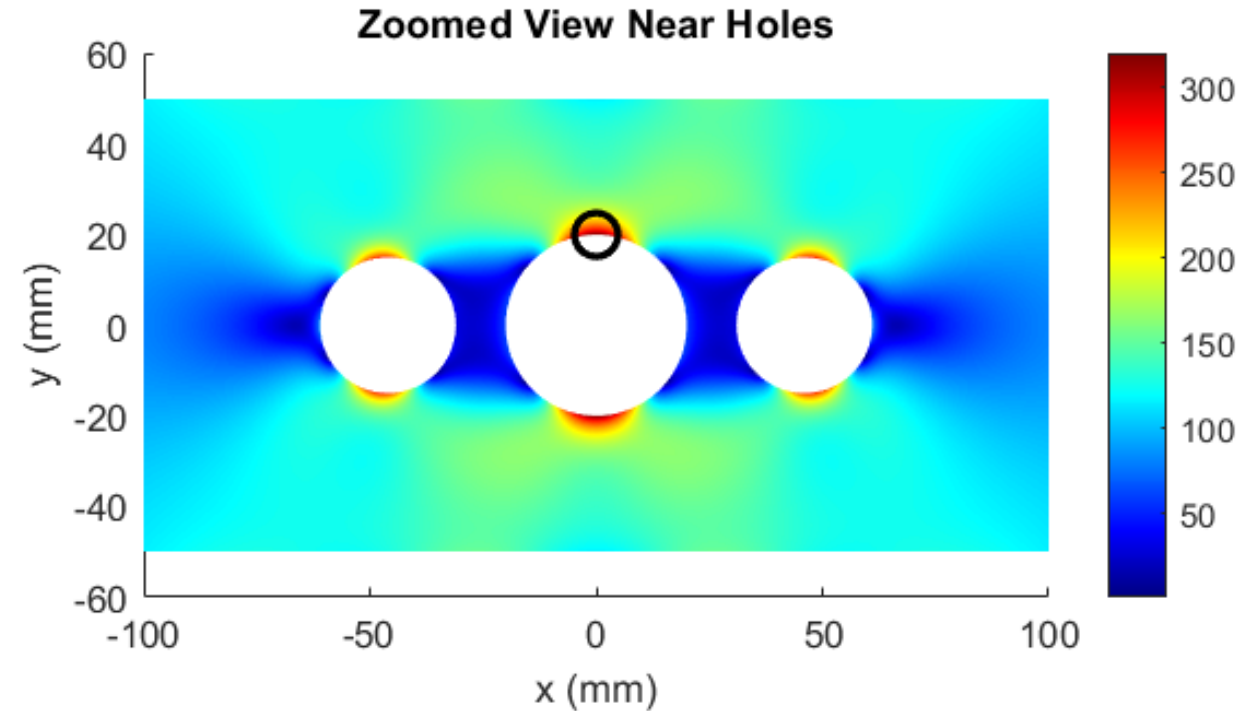


Detailed Stress Analysis

- Maximum stress remains at central hole boundary
- Auxiliary holes redistribute away from critical region
- 11% reduction achieved with simple circular holes

Why circular?

- Uniform stress distribution around perimeter
- No sharp features - Ellipse tips create higher local K_t
- Maximum size hit (upper bound)



Fatigue Life Extension

For high-cycle fatigue, the relationship between stress amplitude and fatigue life follows the **Basquin equation** (S-N curve):

$$N_f = C \cdot \sigma^{-m}$$

- N_f = Number of cycles to failure
- σ = Stress amplitude
- m = Fatigue exponent (material-dependent)
- C = Material constant

The fatigue life improvement factor when stress is reduced:

$$\frac{N_{optimal}}{N_{baseline}} = \left(\frac{\sigma_{baseline}}{\sigma_{optimal}} \right)^m$$

Using the stress reduction from 360.28 to 320.46 Mpa

$$\text{Stress Ratio} = \frac{360.28}{320.46} = 1.124$$

For typical steel component with $m \approx 5$:

- Fatigue life increases by approximately **79%**

If the baseline component fails at 100,000 cycles:

- With auxiliary holes: Expected life $\approx$ **179,000 cycles**

Conclusions

1. Auxiliary holes can reduced stress concentration by 11%
2. Optimal shape was circular for uniaxial tension
3. Fatigue life improvement
4. Multi-start optimization essential due to multiple local minima

Limitations & Future work:

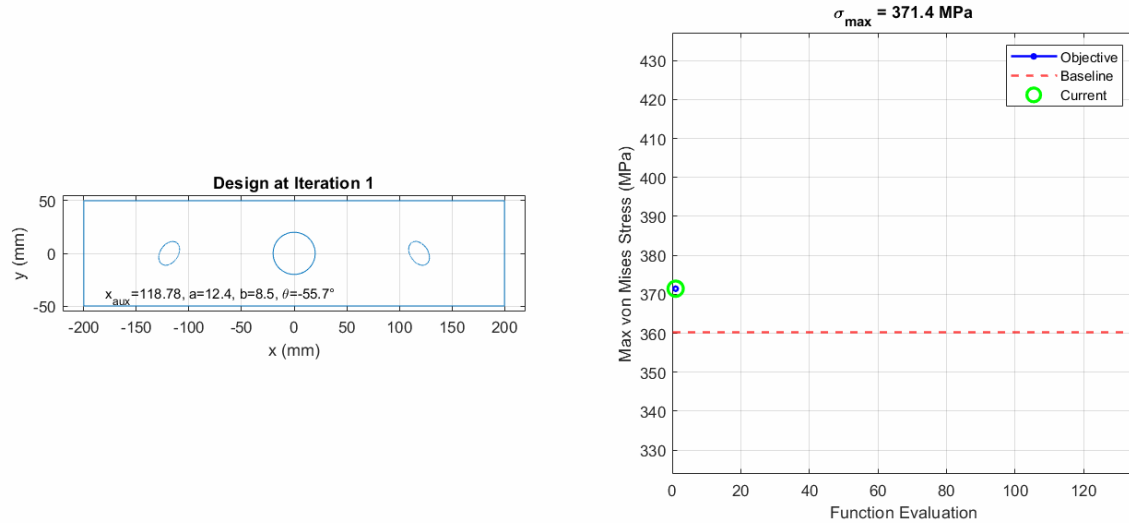
1. 2D analysis
2. Single load case
3. Could explore more auxiliary holes

Extra Slides

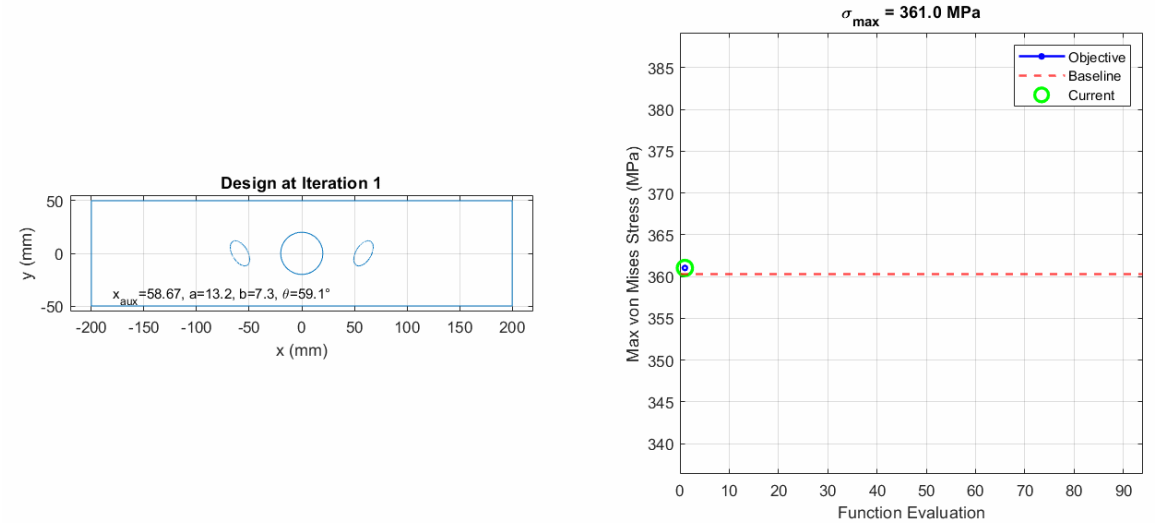


Animation of Runs

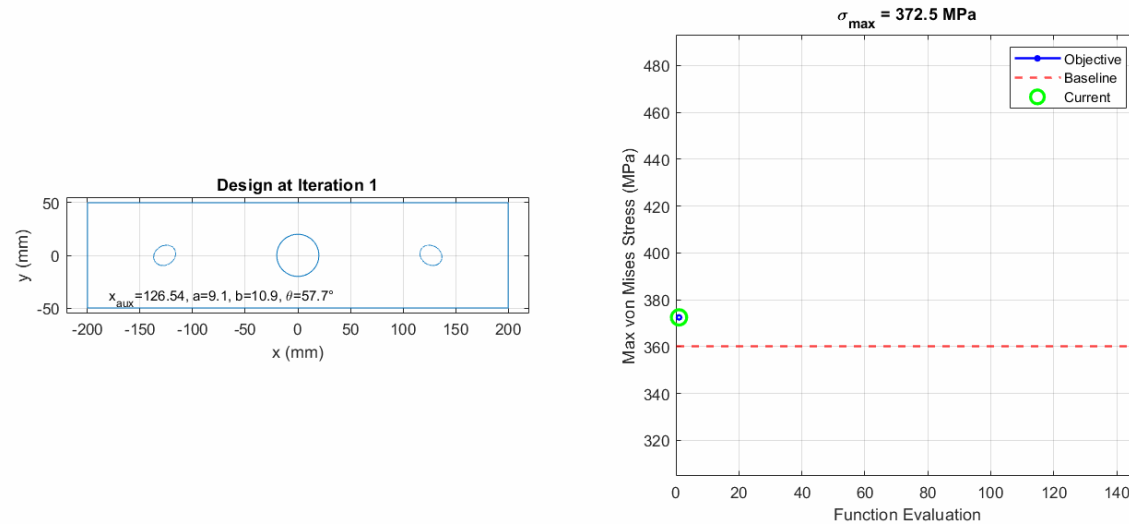
Run 150 - Frame 1/40



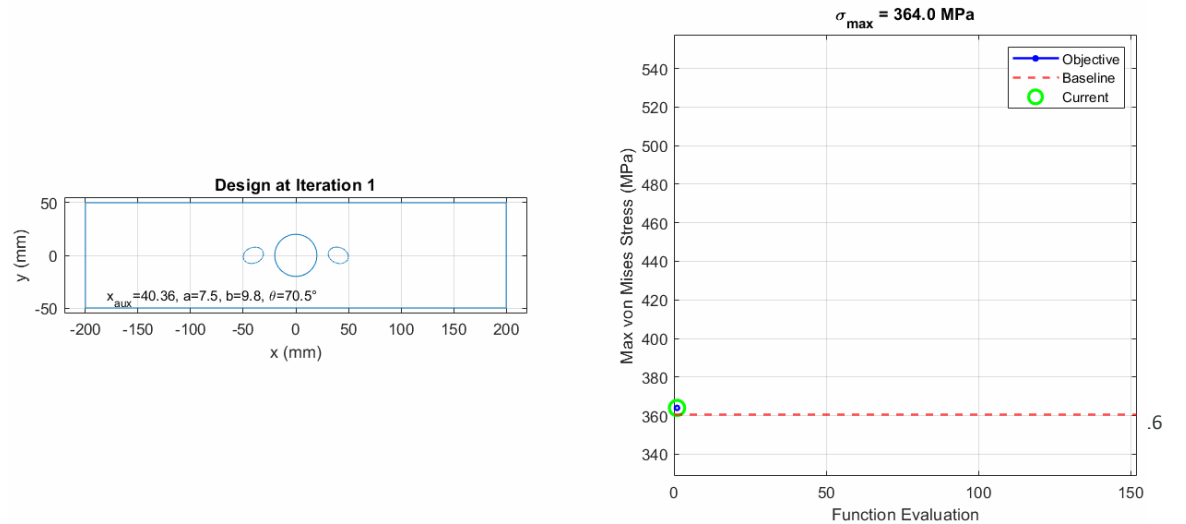
Run 48 - Frame 1/40



Run 19 - Frame 1/40

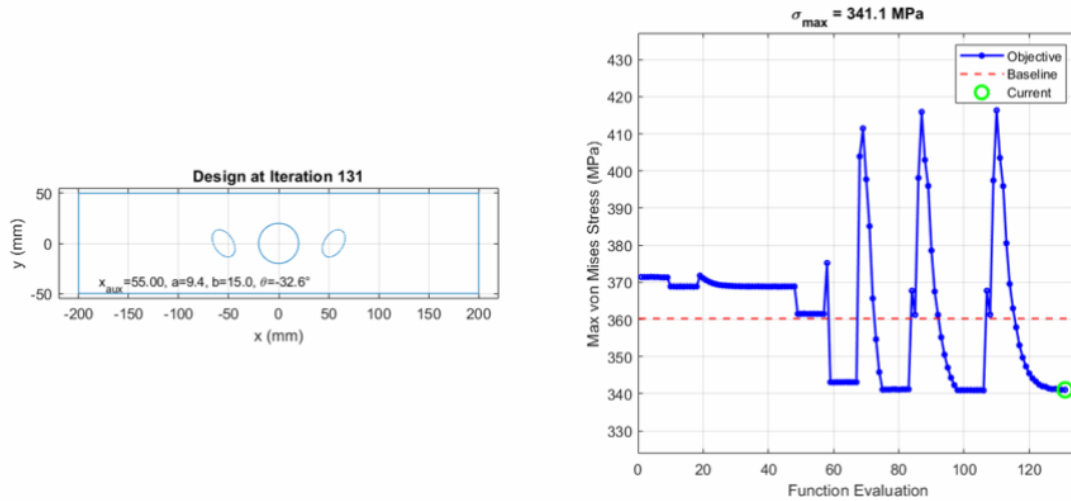


Run 138 - Frame 1/40

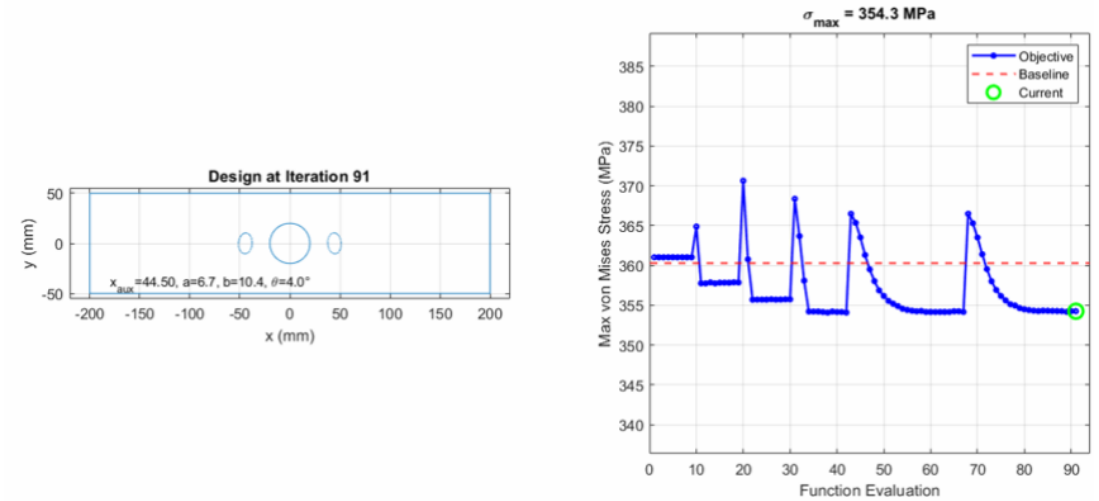


Animation of Runs (Final, converged shape)

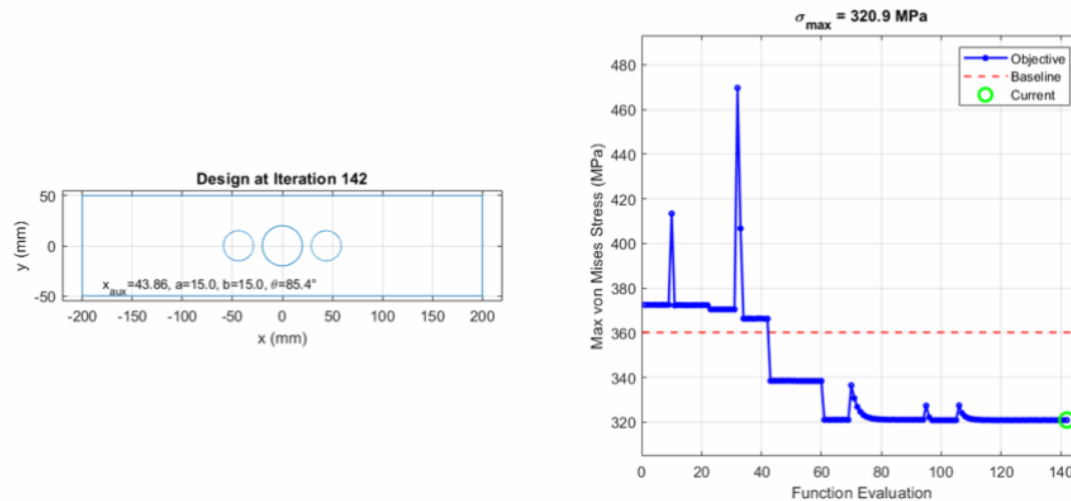
Run 150 - Frame 39/40



Run 48 - Frame 39/40



Run 19 - Frame 39/40



Run 138 - Frame 39/40

